Middleware Architectures 2

Lecture 3: Security

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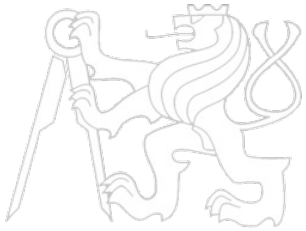

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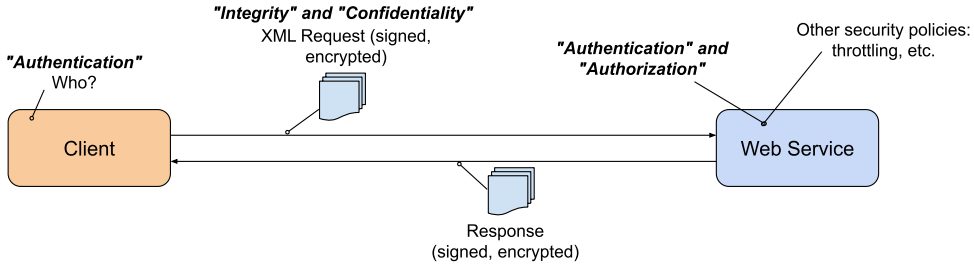
Overview

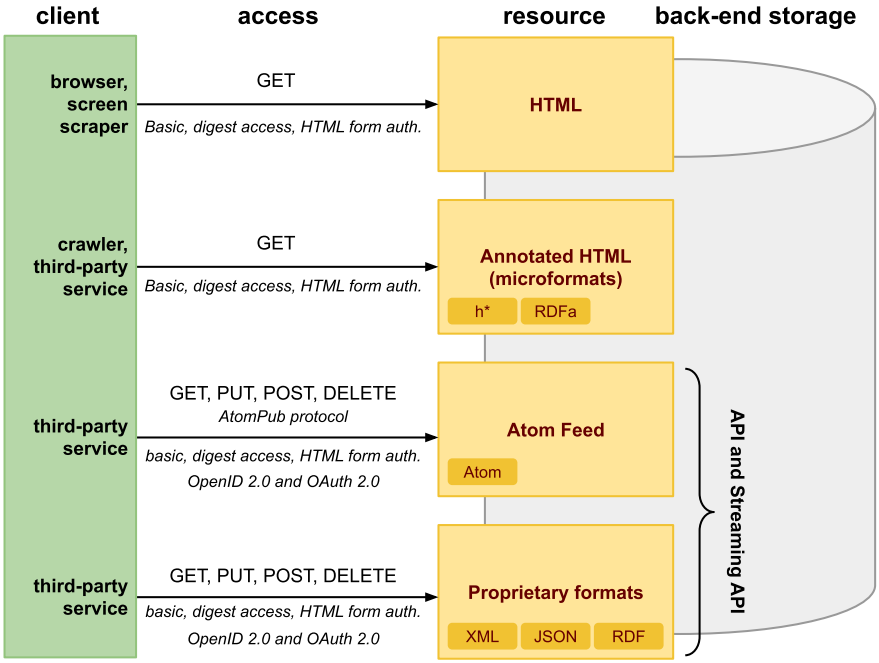
- Security Concepts
  - *Access and Refresh Tokens*
  - *Authorization Models*
- Transport Level Security
- JSON Web Token
- OAuth 2.0
- OpenID

Web Service Security Concepts

- Securing the client-server communcation
  - *Message-level security*
  - *Transport-level security*
- Ensure
  - *Authentication – verify a client's identity*
  - *Authorizaton – rights to access resources*
  - *Message Confidentiality – keep message content secret*
  - *Message Integrity – message content does not change during transmission*
  - *Non-repudiation – proof of integrity and origin of data*



Data on the Web



Authentication and Authorization

- Authentication
  - *verification of user's identity*
- Authorization
  - *verification that a user has rights to access a resource*
- Standard: HTTP authentication
  - *HTTP defines several main options*
    - → *Basic Access Authentication*
    - → *Digest Access Authentication*
    - → *Bearer tokens to access OAuth 2.0-protected resources*
    - → *Mutual authentication using password-based when server knows the user's encrypted password*
  - *Basic and Digest are defined in*
    - → *RFC 2616: Hypertext Transfer Protocol – HTTP/1.1*
    - → *RFC 2617: HTTP Authentication: Basic and Digest Access Authentication*
- Custom/proprietary: use of cookies

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Access Tokens vs Refresh Tokens

- **Access token**
  - *Credential presented to an API/resource server to authorize requests*
  - ***Short-lived** (minutes to ~1 hour): limits damage if leaked*
  - *Sent frequently*
    - → *In an HTTP header (e.g., `Authorization: Bearer`)*
    - → *In a cookie or a request body*
  - *Opaque (generated by the server) or self-contained (e.g., JWT)*
- **Refresh token**
  - *Longer-lived credential used to obtain new access tokens without re-login*
  - *Sent only to the token service / authorization server (not to every API)*
  - *A **high-value secret***
    - → *Closer to a password than an access token*

Token Lifetimes: Security vs Usability

- Why access tokens are short-lived
  - *If an access token leaks, attacker's window is limited by expiry*
  - *Reduces dependency on revocation mechanisms*
- Why refresh tokens exist
  - *Users shouldn't re-authenticate often*
  - *Refresh token enables "silent re-auth" / session continuation*
- Common design pattern
  - *Access token: short TTL, used on every API call*
  - *Refresh token: long TTL, used rarely to mint new access tokens*

Rotation

- **Access token rotation**
  - *Usually "rotates" naturally by minting a new access token when the old one expires*
  - *Old access token may remain valid until expiry (unless you have server-side introspection/revocation)*
- **Refresh token rotation** (one-time use refresh tokens)
  - *Client uses refresh token R1*
  - *Token service returns A2 + R2 and invalidates R1*
- Why rotate refresh tokens
  - *Limits usefulness of a stolen refresh token*
  - *Enables **reuse detection***
    → *using an old refresh token again signals theft/replay*

Refresh Token Reuse Detection

- Threat model
  - *Attacker steals refresh token* `R1`
    → *e.g. malware, XSS, insecure storage, logs*
  - *Both attacker and real client may try to refresh*
- Rotation + reuse detection
  - *Legitimate client refreshes:* `R1`
    → *gets* `R2` *and server invalidates* `R1`
  - *If attacker later submits* `R1` *again*
    → *server detects reuse of an invalidated token*
    → *the whole session can be revoked, force re-authentication, sends alert*

Binding Tokens to Context

- IP binding (soft or strict)
  - *Accept token only if request IP matches expected IP / range*
  - *Tradeoff: reduces replay but can break real users*
    → mobile IP changes, VPNs
  - *Often used as **soft binding***
    → accept token but trigger extra checks
- Device binding / proof-of-possession
  - *Tie token to a device-held key*
  - *Stronger than IP binding but more complex to deploy*
- Client binding
  - *Refresh tokens usable only by the intended client app*
    → Client identity / audience checks

General Protection Mechanisms

- Storage & handling
  - *Protect refresh tokens like passwords*
  - *Never log tokens (server logs, analytics, crash reports)*
  - *Send refresh tokens only to the token service*
- Transport security
  - *HTTPS/TLS everywhere; secure cookie flags*
  - *Consider HSTS*
- Server-side controls
  - *Revocation, rate-limit refresh attempts*
  - *Least privilege scopes/permissions*
    → audience restrictions (token for API A ≠ API B)

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RBAC, ABAC, ACL

- Overall concepts
  - *Triple: subject (user) -- operation -- object (resource)*
- **ACL** (Access Control List)
  - *Each resource stores a list of **who can do what***
    - → *Alice=owner, Bob=read, TeamX=write*
  - *Typical in file systems and document sharing/collaboration tools*
- **RBAC** (Role-Based Access Control)
  - *Users are assigned **roles***
    - → *e.g. admin, editor, viewer*
    - → *roles grant permissions*
  - *Common in enterprise apps and SaaS admin/tenant management*
- **ABAC** (Attribute-Based Access Control)
  - *Decision based on **attributes** of user/resource/context (dept, clearance, time, device, region)*
  - *Large policy-driven organizations*

Role-Based Access Control (RBAC)

- Goal
  - *Control **who can do what** by assigning **roles** to users*
  - *Roles map to a set of **permissions** (operations on resources)*
- Core concepts
  - ***User**: Alice*
  - ***Role**: `admin`, `editor`, `viewer`*
  - ***Permission**: `invoice:read`, `invoice:approve`, `user:invite`*
  - ***Assignment**: Alice has role `editor` ⇒ Alice inherits editor permissions*
- Why RBAC is used
  - *Scales better than per-user ACLs*
  - *Easier audits: "Who are all the admins?"*
  - *Supports least privilege via limited roles*

RBAC Example: Roles → Permissions

- Define roles and permissions mappings

```
viewer:
    project:read
    issue:read
editor:
    project:read
    issue:read
    issue:create
    issue:update
admin:
    project:read
    project:delete
    issue:*
    user:invite
    user:remove
```

- Assign roles to users

```
Alice -> editor
Bob -> viewer
Peter -> (no role)
```

RBAC in an HTTP API (Enforcement)

- Permissions can be checked at the server for each request
  - *Step 1: authenticate user (session cookie / OAuth / JWT)*
  - *Step 2: load roles/permissions*
  - *Step 3: authorize the action (allow/deny)*
- Example API endpoints

```
GET /projects/123 → requires project:read
POST /projects/123/issues → requires issue:create
PUT /projects/123/issues/9 → requires issue:update
POST /org/users/invite → requires user:invite
```

- Example authorization decision
  - *Alice (editor) can* `POST /projects/123/issues`
  - *Bob (viewer) can only read issues, not create/update*
  - *Admins can invite/remove users*

RBAC Shortcomings and Best Practices

- Shortcomings
  - ***Confusing roles with resources****: global* `admin` *vs* `project-admin`
  - *Having permission* `issue:read` *is not enough*
    → *must also check access to issue's project*
- Best practices
  - *Deny by default; keep roles small and explicit*
  - *Use scopes (RBAC + scope):* `role=editor` *in* `project 123`
  - *Log authorization decisions for audits*
  - *Tests*
    → *viewer have read-only access*
    → *editor cannot invite users*

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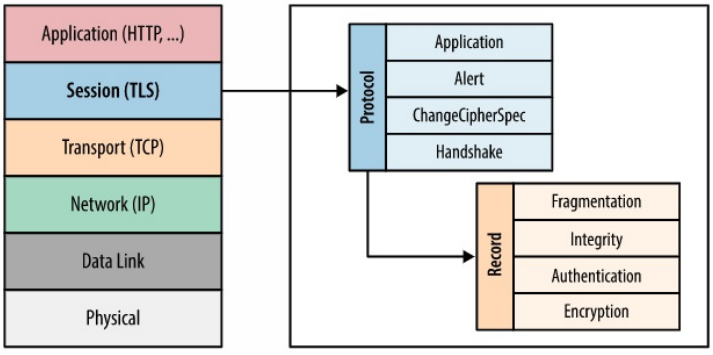
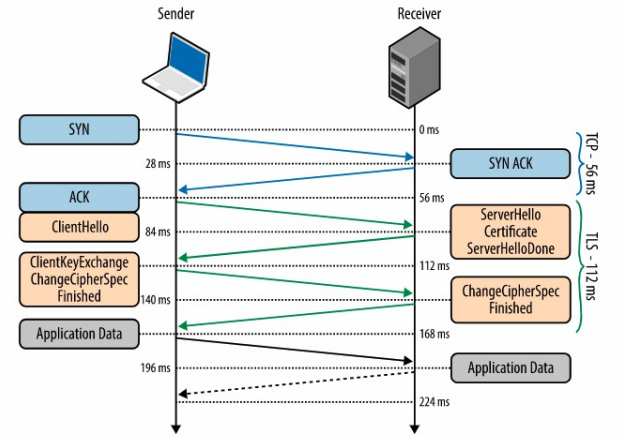
Overview

- SSL and TLS
  - *SSL and TLS is used interchangeably*
  - *SSL 3.0 developed by Netscape*
  - *IETF standardization of SSL 3.0 is TLS 1.0*
    → TLS 1.0 is upgrade of SSL 3.0
  - *Due to security flaws in TLS 1.0, TLS 1.1 and TLS 1.2 were created*
- TLS layer



TLS Handshake Protocol



- TLS Handshake
  56 ms: ClientHello, TLS protocol version, list of ciphersuites, TLS options
  84 ms: ServerHello, TLS protocol version, ciphersuite, certificate
  112 ms: RSA or Diffie-Hellman key exchange
  140 ms: Message integrity checks, sends encrypted "Finished" message
  168 ms: Decrypts the message, app data can be sent

TLS and Proxy Servers

- TLS Offloading
  – *Inbound TLS connection, plain outbound connection*
  – *Proxy can inspect messages*
- TLS Bridging
  – *Inbound TLS connection, new outbound TLS connection*
  – *Proxy can inspect messages*
- End-to-End TLS (TLS pass-through)
  – *TLS connection is passed-through the proxy*
  – *Proxy cannot inspect messages*
- Load balancer
  – *Can use TLS offloading or TLS bridging*
  – *Can use TLS pass-through with help of Server Name Indication (SNI)*

HTTP Strict Transport Security

- Problem
  - *Initial `http://` request (or insecure redirect) enables SSL-stripping*
  - *Mixed links/bookmarks can accidentally use HTTP even if site supports HTTPS*
- Solution
  - *HSTS: HTTP Strict Transport Security*
  - *Server tells browser: "Always use HTTPS for this site for N seconds"*
  - *Browser automatically upgrades future requests to HTTPS before sending traffic*
- Header

```
1 | Strict-Transport-Security: max-age=31536000; includeSubDomains; preload
```

- Parameters
  - `max-age`*: how long the rule is cached (seconds)*
  - `includeSubDomains`*: apply to all subdomains*
  - `preload`*: submit domain to browser preload list (must meet requirements)*

SSL-Stripping

- HTTPS Downgrade Attack
- Eve sits on public Wi-Fi, compromised router, malicious proxy
- Attack flow
  - *Alice requests `http://example.com`*
  - *Server responds with 301/302 redirect to `https://example.com`*
  - *Eve intercepts and modifies the response*
    - → *Removes the redirect*
    - → *Rewrites links to stay http://...*
    - → *Uses HTTPS to server but serves HTTP to victim*
- Impact
  - *Victim remains on **HTTP** while page appears "normal"*
  - *Credentials/actions can be **observed and modified** by attacker*
- Mitigation
  - ***HSTS** (and HSTS preload) forces HTTPS before any request is sent*

Mutual TLS

- Standard TLS
  - *Client authenticates the **server** via server certificate (prevents MITM)*
  - *Client typically authenticates separately (password, cookie, token)*
- Mutual TLS (mTLS)
  - *Both sides authenticate:*
    - *→ Server presents a **server certificate***
    - *→ Client presents a **client certificate** (X.509) during the handshake*
  - *Server verifies client certificate against a trusted CA and policy*
  - *Result: strong **client identity at the transport layer***
- Where it's used
  - *Service-to-service in microservices / service mesh*
  - *Enterprise VPN-less "zero trust" access / internal APIs*
  - *High-trust partner integrations (B2B)*

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- JSON Web Token (JWT)
  - *Open standard (RFC 7519)*
  - *Mechanism to securely transmit information between parties as a JSON object.*
  - *Can be **verified** and **trusted** as it is **digitally signed**.*
- Basic concepts
  - *Compact*
    - *→ has a small size*
    - *→ can be transmitted via a URL, POST, HTTP header.*
  - *Self-contained*
    - *→ payload contains all required user information.*

Use of JWT

- Authentication
  - *After user logs in, following requests contain JWT token.*
  - *Single Sign On widely uses JWT nowadays*
- Information Exchange
  - *Signature ensures senders are who they say they are.*
  - *Message integrity – signature calculated using the header and the payload.*

JWT Structure

`<header>.<payload>.<signature>`

- Header
  - *Contains two parts, the type of the token (JWT) and the hashing algorithm being used (e.g. HMAC, SHA256, RSA).*

```
{
  "alg": "HS256",
  "typ": "JWT"
}
```

- Payload
  - *Contains the claims, i.e. statements about an entity (e.g. user).*
  - *Can be registered, public and private*
  - *Registered and public should be defined in IANA JSON Web Token Registry*

```
{
  "sub": "1234567890",
  "name": "John Doe",
  "admin": true
}
```


JWT Structure (Cont.)

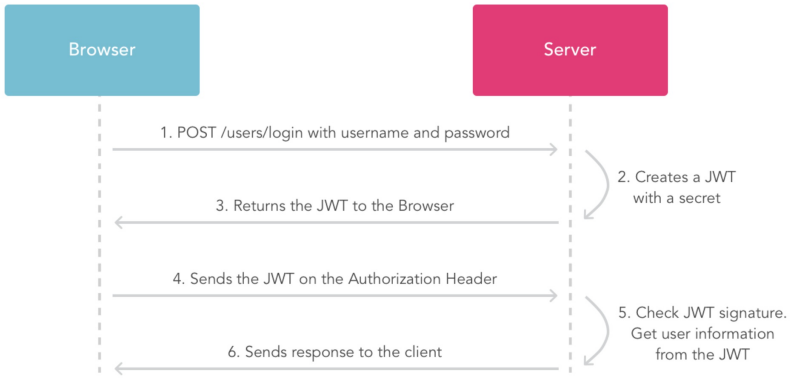
- Signature
  - *Signed encoded header, encoded payload and a secret.*
  - *For example, signature using HMAC SHA256 algorithm*

```
HMACSHA256(
    base64UrlEncode(header) + "." +
    base64UrlEncode(payload),
    secret)
```

- Example
  - *JWT is a three Base64-URL strings separated by dots*

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJzdWIiOiIxMjM0NTY3ODkwIiwibmFtZSI6IkpvaG4
gRG9lIiwiaXNTb2NpYWwiOnRydWV9.
4pcPyMD09olPSyXnrXCjTwXyr4BsezdI1AVTmud2fU4
```

How to use JWT



1. *User sends username and password*
2. *Server verifies user, creates JWT token with a secret and a expiration time*
3. *Server sends JWT token back to the Browser*
4. *Browser sends JWT token on subsequent interations*

Notes

→ Authorization header does not behave the same as cookies!

→ JWT should not contain secrets (passwords) as it can be read (on the client or if non-https connection is used)

Standard Claims and Server-side Validation

- Common registered claims (RFC 7519)
  - `iss` *(issuer): who issued the token*
  - `aud` *(audience): who the token is intended for (which API)*
  - `sub` *(subject): the user / entity identifier*
  - `exp` *(expiration): token must be rejected after this time*
  - `nbf` *(not before): token not valid before this time*
  - `iat` *(issued at): when the token was minted (useful for debugging / policies)*
  - `jti` *(JWT ID): unique identifier (useful for revocation lists)*
- Server-side validation (do this on every request)
  - *Verify the **signature** with the correct key*
    → do not trust the token header blindly
  - *Check* `exp` *(and allow small clock skew)*
  - *Check* `iss` *matches expected issuer*
  - *Check* `aud` *matches your service/API*
  - *If used, enforce* `nbf`/`iat`*; optionally check* `jti` *against a denylist*

Expiration and revocation

- Expiration
  - *Tokens should be valid for a limited time*
  - *Use* `exp` *claim*
    - → *Timestamp of the token expiration*
    - → *Token should be checked on every request*
- Revocation
  - *Tokens (access tokens) usually stored in memory*
  - *Tokens should be refreshed using refresh tokens*
    - → *Refresh tokens are stored in the DB*
    - → *When you need to revoke access tokens stored them in the DB*
    - → *You can expire all tokens by changing the secret on the server*

JWT Security Notes

- **JWT is usually signed, not encrypted**
  - *Base64URL-encoded payload is readable by anyone who has the token*
  - *Do not put secrets (passwords/API keys) in JWTs*
- **Algorithm handling**
  - *Do not accept* `alg: none`
  - *Do not "trust the header" to choose any algorithm*
    - → *allowlist expected algorithms per issuer*
- **Browser storage tradeoffs**
  - *Tokens in JS-accessible storage (e.g., localStorage)*
    - → *vulnerable to XSS theft*
  - *Cookies reduce token theft but require CSRF defenses*
    - → *SameSite/CSRF tokens*
- **Size and performance**
  - *Large JWTs increase request size and can hit proxy/header limits*
    - → *keep claims minimal*

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  - *Client-side Web Apps*
  - *Server-side Web Apps*
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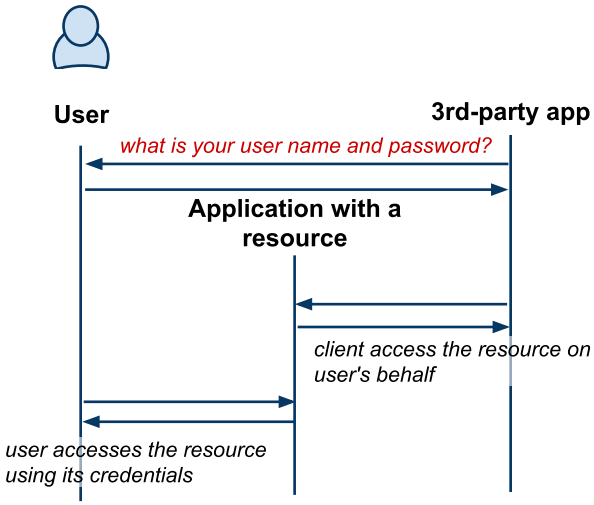
Motivation

- Cloud Computing – Software as a Service
  - *Users utilize apps in clouds*
    - → *they access **resources** via Web browsers*
    - → *they store their data in the cloud*
    - → *Google Docs, Contacts, etc.*
  - *The trend is that SaaS are open*
    - → *can be extended by 3rd-party developers through APIs*
    - → *attract more users ⇒ increases value of apps*
  - *Apps extensions need to have an access to users' data*
- Need for a new mechanism to access resources
  - ***Users can grant access to third-party apps without exposing their users' credentials***

When there is no OAuth



- Users must share their credentials with the 3rd-party app
- Users cannot control what and how long the app can access resources
- Users must trust the app
  - *In case of misuse, users can only change their passwords*

OAuth 2.0 Protocol

- OAuth Objectives
  - *users can grant access to third-party applications*
  - *users can revoke access any time*
  - *supports:*
    - → *client-side web apps (Authorization Code + PKCE),*
    - → *server-side apps (Authorization Code), and*
    - → *native (desktop) apps (Authorization Code + PKCE)*
- History
  - *Initiated by Google, Twitter, Yahoo!*
  - *Different, non-standard protocols first: ClientLogin, AuthSub*
  - *OAuth 1.0 – first standard, security problems, quite complex*
  - *OAuth 2.0 – new version, not backward compatibile with 1.0*
- Specifications and adoption
  - *OAuth 2.0 Protocol*
  - *OAuth 2.0 Google Support*

Terminology

- **Client**
  - *a third-party app accessing resources owned by **resource owner***
- **Resource Owner** (also user)
  - *a person that owns a resource stored in the **resource server***
- **Authorization and Token Endpoints**
  - *endpoints provided by an **authorization server** through which a **resource owner** authorizes requests.*
- **Resource Server**
  - *an app that stores resources owned by a **resource owner***
  - *For example, contacts in Google Contacts*
- **Authorization Code**
  - *a code that a **client** uses to request **access tokens** to access resources*
- **Access Token**
  - *a code that a **client** uses to access resources*

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Client-side Web Apps (SPA)

- Typical scenario
  - *JavaScript/AJAX app running in a browser (public client)*
    → *cannot keep a client secret*
    → *Uses redirect-based login/consent via the user-agent*
- Architecture
  - *User-agent runs the HTML/JS client*
  - *Use **Authorization Code** flow with **PKCE** (recommended for SPAs)*
    → *prevents intercepted* `code` *from being exchanged by an attacker*
- Basic Steps (high level)
  1. *Client redirects user-agent to the authorization endpoint*
     - `response_type=code`, `code_challenge`, `state`)
  2. *Resource owner authenticates + grants access (or rejects)*
  3. *Authorization server redirects back with an **authorization code***
  4. *Client exchanges* `code` + `code_verifier` *for an* `access_token` *(token endpoint)*
  5. *Client calls APIs using* `Authorization: Bearer <access_token>`
  6. *When the access token expires, client obtains a new one (refresh token or re-auth, depending on design)*

PKCE: Proof Key for Code Exchange

- Goal
  - *Prevents an attacker from exchanging a stolen authorization code*
  - *Important for public clients (SPAs, mobile/desktop apps)*
    → *They cannot keep* `client_secret`
- Key idea
  - *Client creates a one-time secret* `code_verifier`
  - *Client sends only a derived value* `code_challenge`
  - *Client later proves possession by sending* `code_verifier`

Step 1: Redirect to Authorization Endpoint

- Goal
  - *Send the user-agent to the **authorization server** to start authentication + consent*
  - *Client includes parameters that bind the response to this browser session (`state`) and enable PKCE*
- Key parameters
  - `response_type=code` – *request an authorization code (not an access token)*
  - `client_id` – *identifies the SPA (public client)*
  - `redirect_uri` – *where the auth server redirects back after login/consent*
  - `scope` – *requested permissions (least privilege)*
  - `state` – *CSRF protection + correlates request/response*
  - `code_challenge` + `code_challenge_method=S256` – *PKCE (binds future token exchange)*
    - → `code_verifier = BASE64URL_ENCODE(secure_random_bytes(32))`
    - → `code_challenge = BASE64URL_ENCODE( SHA256( ASCII(code_verifier) ) )`
- Example request

```
GET https://auth.example.com/oauth2/authorize?
response_type=code&
client_id=spa-client-123&
redirect_uri=https%3A%2F%2Fapp.mydomain.com%2Fcallback&
scope=openid%20profile%20email%20contacts.read&state=af0ifjsldkj
```


Step 3: Redirect Back with Authorization Code

- What happens
  - *After the user authenticates and approves requested scopes, the authorization server redirects the browser to the registered `redirect_uri`*
  - *The redirect includes an authorization code (short-lived, one-time use) and the original `state`*
- Example success callback

```
HTTP/1.1 302 Found
Location: https://app.mydomain.com/callback?
code=SplxlOBeZQQYbYS6WxSbIA&
state=af0ifjsldkj
```

- Client (SPA) responsibilities
  - *Verify `state` matches what was stored before the redirect (CSRF protection)*
  - *Extract `code` and proceed to Step 4 (token endpoint exchange using `code_verifier`)*
  - *Authorization code is not an access token*
- Error callback (user rejects or request invalid)

```
Location: https://app.mydomain.com/callback?
error=access_denied&
state=af0ifjsldkj
```

Step 4: Exchange Code for Tokens

- Goal
  - *Client exchanges the **authorization code** for an* `access_token` *at the token endpoint*
  - ***PKCE** binds the exchange to the original browser session using* `code_verifier`
- Request
  - `POST` *to the token endpoint*
  - *No* `client_secret` *for a SPA*

```
POST /oauth2/token HTTP/1.1
Host: auth.example.com
Content-Type: application/x-www-form-urlencoded
grant_type=authorization_code&
client_id=spa-client-123&
redirect_uri=https%3A%2F%2Fapp.mydomain.com%2Fcallback&
code=SplxlOBeZQQYbYS6WxSbIA&
code_verifier=dBjftJeZ4CVP-mB92K27uhbUJU1p1r_wW1gFWFOEjXk
```

- Response (example)

```
{ "access_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer", "expires_in": 3600,
  "scope": "openid profile email contacts.read" }
```

- Security notes
  - `code` *is short-lived and single-use*
  - *If an attacker steals the* `code` *but not the* `code_verifier`*, the exchange fails*
  - *Tokens should be sent only over HTTPS; never put tokens in URL query*

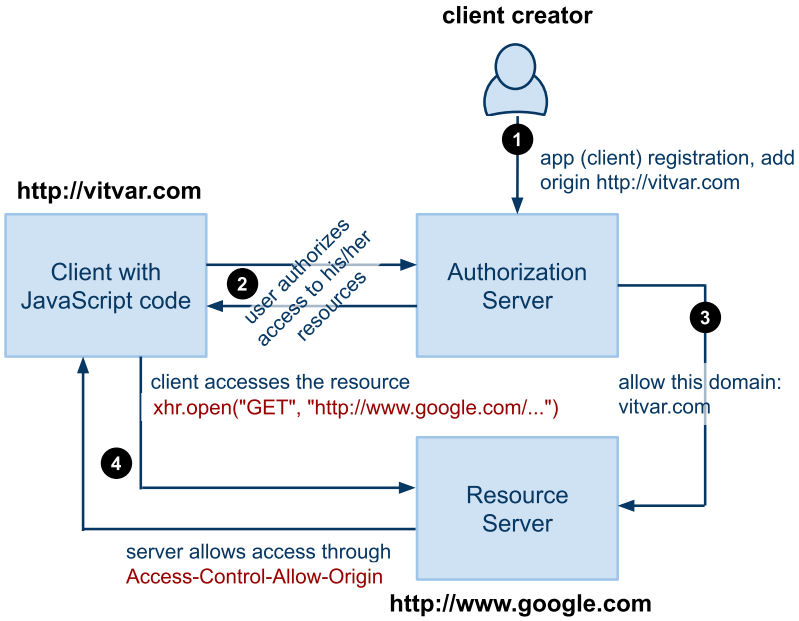
Step 5: Call APIs with Access Token

- Goal
  - *Use the* `access_token` *to access protected resources on the **resource server***
  - *Send tokens in the **Authorization** header (avoid URL query parameters)*
- HTTP request (recommended)

```
GET /v1/contacts HTTP/1.1
Host: api.example.com Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

- What the resource server does
  - *Validates the token (signature,* `exp`*,* `iss`*,* `aud`*, scopes)*
  - *Enforces permissions based on* `scope` */ claims*
- Common responses
  - *Success:* `200 OK`
  - *Missing/invalid token:* `401 Unauthorized`
  - *Valid token but insufficient scope/permission:* `403 Forbidden`

Cross-Origin Resource Sharing



– *see Same Origin and Cross-Origin for details*

Example Application Registration

Google apis

vitvar.com search

- Overview
- Services
- Team
- API Access
- Billing
- Reports
- Quotas

API Access

To prevent abuse, Google places limits on API requests. Using a valid OAuth token or API key allows you to exceed anonymous limits by connecting requests back to your project.

Authorized API Access

OAuth allows users to share specific data with you (for example, contact lists) while keeping their usernames, passwords, and other information private. Learn more

Branding information

The following information is shown to users whenever you request access to their private data.

Product name: w20-test

Google account: t.vitvar@gmail.com

Edit branding information...

Client ID for web applications

Client ID: 621535099260.apps.googleusercontent.com

Client secret: RxWM9l7Sv-7cyfWMW7KhNV9R

Redirect URIs: http://vitvar.com/examples/oauth/callback.html

JavaScript origins: http://example.org

Edit settings...

Reset client secret...

Create another client ID...

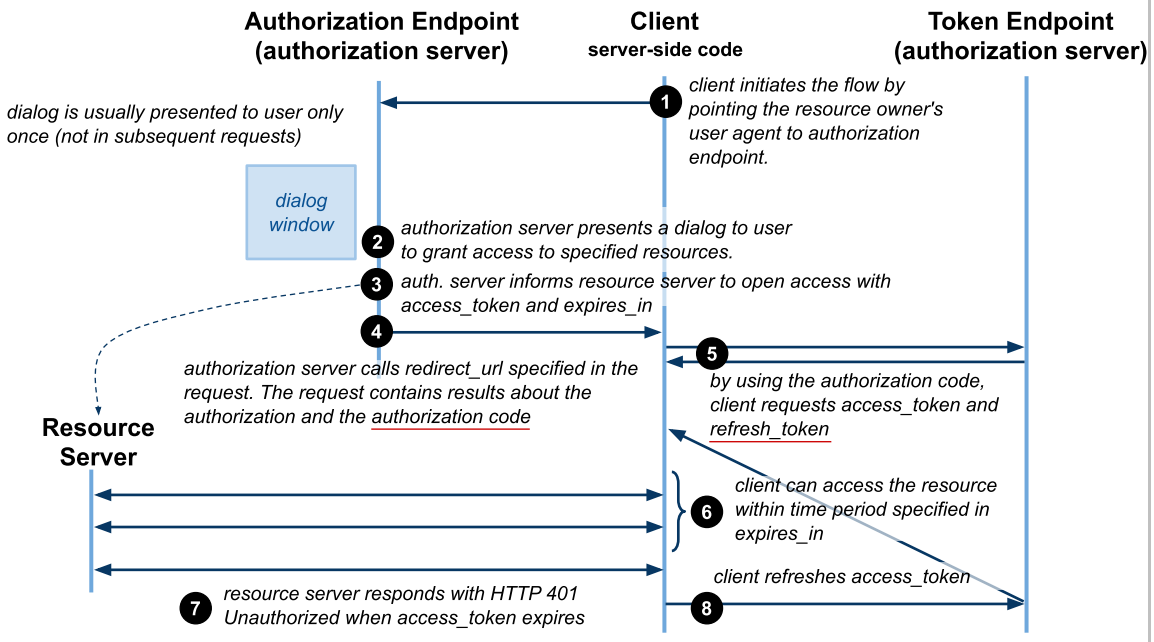
Overview

- Security Concepts
- Transport Level Security
- JSON Web Token
- OAuth 2.0
  - *Client-side Web Apps*
  - *Server-side Web Apps*
- OpenID

Server-side Web Apps

- Additional interactions
  - *server-side code (any language), the app can maintain the state*
  - *additional interactions, authorization code*
- Architecture
  - *Client at a server requests, remembers and refresh access tokens*
- Basic steps
  - *Client redirects user agent to the authorization endpoint*
  - *Resource owner grants access to the client or rejects the request*
  - *Authorization server provides **authorization code** to the client*
  - *Client requests **access** and **refresh tokens** from the auth. server*
  - *Client access the resource with the access token*
  - *When the token expires, client refreshes a token with refresh token*
- Advantages
  - *Access tokens not visible to clients, they are stored at the server*
  - *more secure, clients need to authenticate before they can get tokens*

Server-side Web Apps Protocol



Redirection – Step 1

- Methods and Parameters
  - *same as for client-side app, except* `response_type` *must be* `code`
- Example

```
https://accounts.google.com/o/oauth2/auth?
client_id=621535099260.apps.googleusercontent.com&
redirect_uri=https://w20.vitvar.com/examples/oauth/callback.html&
scope=https://www.google.com/m8/feeds&
state=af0ifjsldkj&
response_type=code
```

Callback + Access Token Request – steps 4, 5

- Callback
  - *authorization server calls back* `redirect_uri`
  - *client gets the* `code` *and requests* `access_token`
  - *client verifies the* `state` *parameter*
  - *example (resource owner grants access):*
    `https://w20.vitvar.com/examples/oauth/callback.html?code=4/P7...`
  - *when user rejects → same as client-side access*
- Access token request
  - `POST` *request to token endpoint*
    → example Google token endpoint:
    `https://accounts.google.com/o/oauth2/token`

```
1  POST /o/oauth2/token HTTP/1.1
2  Host: accounts.google.com
3  Content-Type: application/x-www-form-urlencoded
4
5  code=4/P7q7W91a-oMsCeLvIaQm6bTrgtp6&
6  client_id=621535099260.apps.googleusercontent.com&
7  client_secret=XTHhXh1S2UggvyWGwDk1EjXB&
8  redirect_uri=https://w20.vitvar.com/examples/oauth/callback.html&
9  grant_type=authorization_code
```


Access Token (cont.)

- Access token response
  - *Token endpoint responds with* `access_token` *and* `refresh_token`

```
1  { "access_token"   : "1/fFAGRNJru1FTz70BzhT3Zg",
2    "expires_in"     : 3920,
3    "refresh_token"  : "1/6BMfW9j53gdGImsixUH6kU5RsR4zwI9lUVX-tqf8JXQ" }
```

- Refreshing a token
  - `POST` *request to the token endpoint with* `grant_type=refresh_token` *and the previously obtained value of* `refresh_token`

```
1  POST /o/oauth2/token HTTP/1.1
2  Host: accounts.google.com
3  Content-Type: application/x-www-form-urlencoded
4
5  client_id=21302922996.apps.googleusercontent.com&
6  client_secret=XTHhXh1SlUNgvyWGwDk1EjXB&
7  refresh_token=1/6BMfW9j53gdGImsixUH6kU5RsR4zwI9lUVX-tqf8JXQ&
8  grant_type=refresh_token
```

- Accessing a resource is the same as in the client-side app

Overview

- Security Concepts
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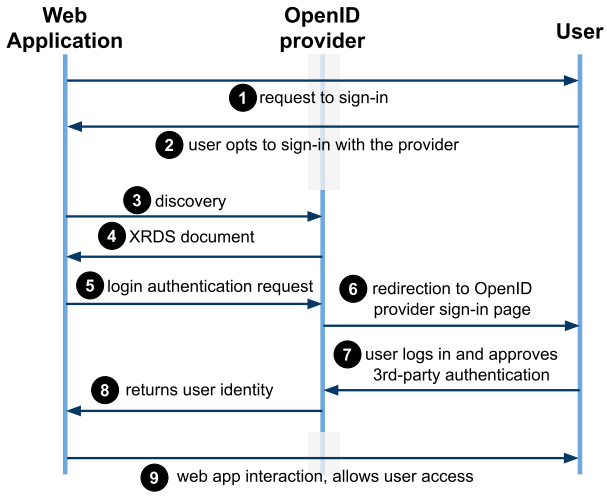
- OpenID
  - *OpenID Connect*

OpenID Protocol

- Motivation – many user accounts
  - *users need to maintain many accounts to access various services*
  - *multiple passwords problem*
- Objectives
  - *allows apps to utilize an OpenID provider*
    - → *a third-party authentication service*
    - → *federated login*
  - *users have one account with the OpenID provider and use it for apps that support the provider*
- OpenID providers
  - *it is a protocol, anybody can build a provider*
  - *Google, Yahoo!, Seznam.cz, etc.*
- Specification
  - *OpenID Protocol*

Interaction Sequence



- Discovery – discovery of a service associated with a resource
- XRDS – eXtensible Resource Descriptor Sequence
  - *format for discovery result*
  - *developed to serve resource discovery for OpenID*
  - *Web app retrieves endpoint to send login authentication requests*

Login Authentication Request – Step 5

- Example Google OpenID provider

```
https://www.google.com/accounts/o8/id
?openid.ns=http://specs.openid.net/auth/2.0
&openid.return_to=https://www.example.com/checkauth
&openid.realm=http://www.example.com/
&openid.assoc_handle=ABSmpf6DNMw
&openid.mode=checkid_setup
```

- Parameters
  - `ns` – *protocol version (obtained from the XRDS)*
  - `mode` – *type of message or additional semantics (*`checkid_setup` *indicates that interaction between the provider and the user is allowed during authentication)*
  - `return_to` – *callback page the provider sends the result*
  - `realm` – *domain the user will trust, consistent with* `return_to`
  - `assoc_handle` – *"log in" for web app with openid provider*

  ** Not all fields shown, check the OpenID spec for the full list of fields and their values*

Login Authentication Response – Step 8

- User logins successfully

```
http://www.example.com/checkauth
?openid.ns=http://specs.openid.net/auth/2.0
&openid.mode=id_res
&openid.return_to=http://www.example.com:8080/checkauth
&openid.assoc_handle=ABSmpf6DNMw
&openid.identity=https://www.google.com/accounts/o8/id/id=ACyQatiscWvwqs4UQV_l
```

  - *Web app will use* `identity` *to identify user in the application*
  - *response is also signed using a list of fields in the response (not shown in the listing)*

- User cancels

```
http://www.example.com/checkauth
?openid.mode=cancel
&openid.ns=http://specs.openid.net/auth/2.0
```

** Not all fields shown, check the OpenID spec for the full list of fields and their values*

Overview

- Security Concepts
- Transport Level Security
- JSON Web Token
- OAuth 2.0
- OpenID
  - *OpenID Connect*

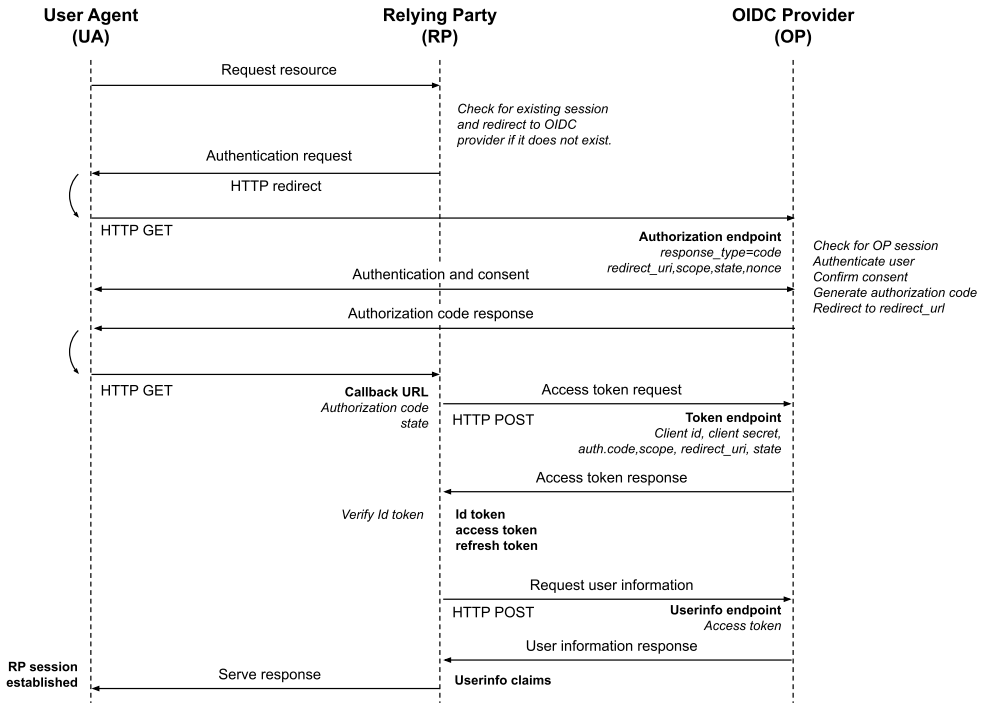
OpenID Connect (OIDC)

- Simple identity layer on top of the OAuth 2.0 protocol
  - *Authorization Server to verify identity of users*
  - *Clients can obtain basic profile information about users*
  - *OAuth answers: "what can this app access?"*
  - *OIDC answers: "who is the user?"*
- OIDC vs OpenID
  - *OIDC does many of the same tasks as OpenID 2.0*
  - *API-friendly*
    → *can be used by native and mobile applications*
  - *Robust signing and encryption mechanisms*
  - *Native integration with OAuth 2.0.*
- Defined by OpenID open standard
  - *OpenID Connect*

OIDC Roles and Tokens

- Roles (same actors as OAuth, plus OIDC specifics)
  - ***End-User**: the user who is logging in*
  - ***Relying Party (RP)**: the client application (OAuth client)*
  - ***OpenID Provider (OP)**: the identity provider*
    → *OAuth authorization server*
- Tokens
  - ***ID Token** (new in OIDC): proves authentication; contains identity claims*
    → *almost always a **JWT** signed by the OP*
  - ***Access Token** (OAuth): used to call APIs / resource server*
  - ***Refresh Token** (optional): used to get new tokens without re-auth*

Interaction sequence



OIDC Step 1: Authentication Request

- **Same redirect as OAuth**, but request includes OIDC parameters
  - `scope` *must include* `openid` *(otherwise it is just OAuth)*
  - `nonce` *binds the ID Token to this browser session (replay protection)*
  - `state` *protects against CSRF and correlates request/response*
  - *For SPAs/native apps*
    → *use Authorization Code + PKCE*
- Example (Auth Code + PKCE)

```
GET https://idp.example.com/authorize?
response_type=code&
client_id=rp-client-123&
redirect_uri=https%3A%2F%2Fapp.mydomain.com%2Fcallback&
scope=openid%20profile%20email&
state=af0ifjsldkj&
nonce=n-0S6_WzA2Mj&
code_challenge=E9Melhoa2OwvFrEMTJguCHaoeK1t8URWbuGJSstw-cM&
code_challenge_method=S256
```

OIDC Step 2: User Authentication + Consent

- What happens at the OpenID Provider (OP)
  - *User-agent displays the OP login page (if not already signed in)*
  - *End-User authenticates*
    → *password, MFA, passkey, etc. (handled by the OP)*
  - *OP asks for consent (if needed)*
    → *based on requested* `scope` *(e.g.,* `openid`*,* `profile`*,* `email`*)*
- Important notes
  - *Client never sees the user's password*
  - *OP establishes its own session with the browser (often via cookies)*

  ***Result**: OP is ready to issue an authorization response (next step)*

OIDC Step 3: Redirect Back with Authorization Code

- After successful authentication (and consent), OP redirects back
  - *Redirect target is the registered* `redirect_uri`
  - *Response includes an* `code` *and the original* `state`
- Example success callback

```
HTTP/1.1 302 Found
Location: https://app.mydomain.com/callback?
code=SplxlOBeZQQYbYS6WxSbIA&
state=af0ifjsldkj
```

- Client responsibilities
  - ***Verify*** `state` *matches what was stored in Step 1 (CSRF protection)*
  - *Extract* `code` *and continue to Step 4 (token endpoint exchange)*
- Error callback (user rejects or request invalid)

```
Location: https://app.mydomain.com/callback?
error=access_denied&
state=af0ifjsldkj
```

OIDC Step 4: Token Request

- Client exchanges the authorization `code` at the token endpoint
- Example (server-side)

```
POST /oauth2/token HTTP/1.1
Host: idp.example.com
Content-Type: application/x-www-form-urlencoded
grant_type=authorization_code&
client_id=server-client-123&
client_secret=CLIENT_SECRET&
redirect_uri=https%3A%2F%2Fserver.mydomain.com%2Fcallback&
code=SplxlOBeZQQYbYS6WxSbIA
```

- Token response
  - `access_token` *(OAuth)*
  - `id_token` *(OIDC – identity token)*
  - `refresh_token` *(optional)*
- Example response

```
{ "access_token": "SlAV32hkKG",
  "token_type": "Bearer",
  "expires_in": 3600,
  "id_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9..." }
```


ID Token (JWT) – Typical Claims

- Registered claims to validate
  - `iss` – *expected issuer (the OP)*
  - `aud` – *must contain your* `client_id` *(intended audience)*
  - `exp` – *not expired (allow small clock skew)*
  - `iat` – *issued at (optional policy)*
  - `nonce` – *must match the nonce sent in Step 1 (replay protection)*
- User identity
  - `sub` – *stable identifier for the End-User at this OP*
  - *Profile claims are often requested via scopes*
    → `profile`, `email`, `phone`, `address`

Request User Information (optional)

- Purpose
  - *Retrieve additional user profile claims from the OP's* `UserInfo` *endpoint*
  - *Use when you need more claims than what is in the* `id_token` *(or need fresher data)*
- Important notes
  - *Call* `/userinfo` *using the* ***access token*** *(not the* `id_token`*)*
  - *Returned claims depend on granted scopes (e.g.,* `profile`*,* `email`*)*
- Example request

```
GET /userinfo HTTP/1.1
Host: idp.example.com
Authorization: Bearer access_token
Accept: application/json
```

- Example response

```
{ "sub": "248289761001",
  "name": "Jane Doe",
  "given_name": "Jane",
  "family_name": "Doe",
  "email": "jane.doe@example.com" }
```